

Vuuro Scan: property reality capture, and it is yours to build

Date: 6 August 2026

For: Joven · From: Mark

Status: direction brief and complete kickoff package for this window. No further research packs or architecture dumps will be handed during my low-availability period. Everything you need to start and steer is in this document and the access we give you at kickoff.

One package, then you steer. Treat this brief as the full written brief for the coming period. Do not wait on more docs from me. If something is missing, decide with a short rationale, keep moving, and surface it in the daily channel update.

1. North star

What we are putting on the table, before any technique.

We are going to own property reality capture for Vuuro. A branded scanning product beside the rental platform: floor plans, measurements, photos and notes, bound to Vuuro property, unit and organisation identity from day one, and joined into the rental codebase later over API. Landlords and agents place and enrich listings with real geometry, not gallery dumps alone.

Where this lands in a year or two: scan once on site, and the listing, the inspection record and the renovation baseline all drink from the same capture. Indicative Dutch residential metrics with honest labeling. Optional higher-accuracy paths when someone needs them. A product in its own right, not a sidecar photo tool, and a clear reason landlords prefer listing and operating on Vuuro.

That is a product, not a plumbing job. It will outlive any single phone OS release and any single reconstruction library we put behind it.

GEOMETRY

Floor plans and measurements that belong to the unit

API-READY

Scan Service contract first; Vuuro app wires later

HONEST m²

Indicative metrics, never fake certification

OPS MEMORY

Check-in, check-out, renovation baselines

2. What the product does

The capabilities that define it. How you build them is your call.

- **It captures rooms fast on Pro devices.** Guided LiDAR / RoomPlan-class flows first; clear messaging when a device cannot do it; a designed AR fallback so non-Pro is not a dead end.

- **It attaches to Vuuuro identity.** Every scan session carries property, unit and organisation identifiers. No orphan files on a phone. The rental app will consume those payloads later over API.
- **It produces usable listing assets.** 2D floor plans, room list, indicative areas, photos and notes as stable API-shaped results. Vuuuro web UI wire-up into the rental codebase comes in a later coupling step, not as a week-one blocker.
- **It measures honestly.** NEN2580-inspired numbers are fine for residential intuition. Official certification is a different product. Label the difference in the UI and in exports.
- **It remembers purpose.** Listing capture, check-in, check-out and renovation planning are first-class reasons to scan, not afterthought tags.
- **It respects interiors as personal data.** Consent for occupied units, tenant-scoped storage, no casual public links to home geometry.
- **It leaves room to grow.** Multi-room quality, exports (PDF/PNG first), optional laser pairing, and later Android only when the iOS path has earned it.

Think of Vuuuro Scan as a product with an API and a data model, not a pile of USDZ files in a Drive folder. Build the Scan Service and contract first. The Vuuuro rental codebase couples later as an API client. Listing creation, inspections and maintenance drink from that same contract once we wire them.

3. Why now

The pain is real and current.

- Listings are often photo-heavy and plan-poor. Tenants cannot read the layout; landlords look less professional than the market they compete with.
- Square metres are guessed, copied from old deeds, or left vague. That kills trust and weakens every downstream conversation about rent and space.
- Capture today is fragmented: phone gallery, DIY sketches, external tools that never write back into Vuuuro.
- Competitors in NL already sell LiDAR floor plans as a workflow. Vuuuro wins by owning the integration into rental listing and property ops, not by inventing depth sensors.
- We want landlords to place and rent their own objects on Vuuuro with beauty and speed. Geometry is part of that promise.

4. Terrain already gathered

We did the digging so you can start improving instead of excavating. What follows is the usable terrain.

There is no second document drop planned for this window.

- **iOS Pro + RoomPlan is the practical starting surface.** Depth-backed room capture exists as a productised Apple API; raw ARKit meshes are the heavier path for whole-property work later.
- **Hardware lock-in is real.** LiDAR is Pro-only. Non-Pro and Android need an honest fallback story; do not hard-gate listing creation on owning a Pro phone.
- **Market patterns are settled enough.** Guided capture, room-by-room sessions, photos/notes, cloud post-processing, and multi-format export show up across Immoscan, magicplan, Canvas and Matterport-class tools.

- **NEN2580 is a liability boundary.** Mobile LiDAR can approximate residential metrics. It does not replace a certified survey without a human QC path. Disclaimers are product, not polish.
- **Privacy is not optional.** Interior scans of homes are personal-data-adjacent. Consent, retention and tenant ACL belong in the first design, not in a later cleanup.
- **Integration shape is clear, timing is staged.** Mobile captures and uploads; a Scan Service processes and exposes results over API. The Vuuro rental codebase couples to that API later. Design the contract as if Vuuro were already a client; do not wait on deep rental-app surgery to prove capture.

5. Hard constraints

Eight non-negotiables. Everything not on this list is yours to shape.

- **Identity-native captures.** Every scan session carries property, unit and organisation identifiers in the data model and API. Orphan captures are a bug.
- **Honest measurement language.** Never present mobile LiDAR output as certified NEN2580 unless a certified/QC path actually ran.
- **Privacy by design.** Occupied-unit consent, tenant-scoped storage, audited access. No "anyone with the link" defaults for interior geometry.
- **API-first Vuuro coupling.** Design Scan Service results so the Vuuro rental codebase can consume them over API later. Do not treat deep edits inside the rental app as a prerequisite for proving capture. When we couple, we couple through that contract.
- **Device honesty.** Detect LiDAR / RoomPlan capability. Unsupported devices get a clear message and a designed fallback path, not a crash or a silent empty feature.
- **You own the net under the work.** Not "be more careful". A real independent check that runs without me, built from different assumptions than the code it guards, and whose verdict you cannot merge past. You already built a code-quality agent; point that class of net at this product.
- **Evidence over theatre.** You just send daily updates in the channel. Structure them in three buckets: verified working, implemented but not yet verified, at risk. That split beats any polished status story, and it is how I can stay light-touch for three weeks without losing the truth.
- **No silent production leap.** Local plus CI (and the independent net) prove slices before App Store or production deploy. Store launch is earned, not assumed in week one.

6. The arc ahead

Three movements, not a schedule. Pace and order inside them are yours.

FIRST

Proof of capture

One room scanned on a capable device, uploaded to Scan Service, bound to property/unit/org IDs, with basic dimensions readable end to end through the API contract.

THEN

Unit story

Multi-room session, backend floor plan plus indicative areas, photos and notes, as a coherent unit package served by the Scan Service API.

AFTER THAT

Pilot hardening

Coverage/quality cues, PDF/PNG exports, privacy/ACL polish, optional laser spike, inspection purpose tags. Ready for a small landlord pilot once Vuuro API coupling is live.

No App Store or production server deploy is required to get going. Local proof plus a clean Scan Service API is enough for the first arc. Wiring that API into the Vuuro rental codebase is a later, deliberate coupling step. I would rather you get capture-to-contract right than rush a store listing or a half-done rental-app merge.

This is orientation, not a day plan. If your sequencing beats mine, take yours and tell me why.

7. Decided, open, and out of scope

Decided means settled, do not relitigate without a strong finding. Open means a default exists so you are never blocked, and the better argument wins.

DECIDED

- iOS first; RoomPlan-class room capture is the MVP spine.
- A dedicated Scan Service (bounded context) handles ingest/processing/storage and exposes results over API.
- Vuuro rental-app coupling is later and API-based. This window proves capture and the contract; it does not require deep rental-codebase surgery.
- Listing enrichment and property identity are the first business value, not CAD parity.
- Export priority for early value: floor plan image/PDF and metrics through the Scan Service API before exotic CAD formats.
- No further written packs during my low-availability window. This brief is the kickoff package.

OPEN, DEFAULTS SET

- RoomPlan-only stitching vs early custom ARKit mesh for multi-room.
- How much preview runs on-device vs authoritative metrics on the backend.
- Pure Apple stack vs a specialised reconstruction/export SDK.
- Build-only vs partner/white-label hybrid later; challenge with unit economics and control, not vibes.
- Exact libraries, repo layout, auth shape of the Scan Service API, and the thin demo consumer you use until Vuuro couples.

- Official certified NEN2580 as a product is later, not this arc.
- Android parity and Matterport-class marketing tours are later chapters.
- **SIDE TRACK** Agent / code-quality trial work sits in the background unless we explicitly pull you back. This workstream is the main track.

8. How we will work

This is an ownership role, not a ticket queue. You run Vuuro Scan like it is yours, because it is. Open decisions are yours to take, with a short written rationale so we can follow your thinking. Where I set a default, challenge it if you see better; I will take a good argument over my own default every time. The one thing that does not work is quietly stepping over a hard constraint. If one of them blocks you, say so and we change it together.

The special condition for this workstream is the net. Your measured gap is not speed or depth; it is second-order damage beside the case you were fixing. So ownership here includes an independent check

that does not share the assumptions of the code under test, that you cannot merge past, and that runs without me. Build that once, keep it honest, and use it.

I am less available for about three weeks from next week. You just send daily updates in the channel. Flag blockers the day you hit them. Do not wait on extra material from me. During that period I need to be able to assume the net holds, and that you can keep moving from this brief without waiting on me between updates.

Why you. This product needs someone who can hold an architecture in their head, move fast without waiting for tickets, and turn a hard verification lesson into a permanent working habit. That is the work on the table here, and it compounds. Take it and make it yours - after you answer the open bar question from my last message.

Direction brief for Vuuro Scan (market, RoomPlan/ARKit, NEN2580, privacy, API-first integration). This document is the kickoff contract for the coming window. No companion research PDF will be attached or drip-fed during my low-availability period.